Inurum Enterprise DashCam

TruckerCloud Integration API Reference Guide

Direct Telematics, ADAS/DMS Safety Events, Historical Breadcrumbs, and On-Demand Video Retrieval

Target Audience: TruckerCloud Integration Team	**Production Base URL:** http://3.85.81.10:8080
Auth Header: X-API-KEY: inurum-tc-prod-key-2026	**OpenAPI Docs:** http://3.85.81.10:8080/swagger-ui/index.html
Compliance: 17-char VIN, ECU Odometer, ISO 8601 UTC	**Hardware:** WanWay H20P 4G LTE Dual-Cam (Road + Cabin)

1. System Overview & Architecture

Inurum operates an independent DashCam and telematics platform connecting directly to physical WanWay H20P dual-camera hardware via JT/T 808 (telemetry/commands on port 7800) and JT/T 1078 (RTP media transfer on port 10780). All continuous video footage records locally onto on-device high-endurance MicroSD cards to eliminate unnecessary cloud egress and storage overhead. Selected incident clips (harsh events, collisions, DMS driver alerts) and dispatcher-requested recordings are indexed and made available on-demand through these certified REST APIs.

Standard	Implementation Specification
Vehicle Identification	Strictly enforced 17-character ISO 3779 VIN with valid letters/digits (excluding I, O, Q).
Odometer Data	Always labeled with provenance: ECU (direct vehicle bus) or GPS_ESTIMATED. Units in kilometers.
Time Synchronization	All timestamps are formatted in UTC ISO 8601 (e.g., 2026-11-03T10:30:00Z).
Authentication	Server-to-server HTTP header: X-API-KEY: inurum-tc-prod-key-2026. Global Swagger/API key supported.
Video Streaming	HLS format (m3u8 index with MPEG-TS segments) for both Channel 1 (Road) and Channel 2 (Cabin).

2. Certified TruckerCloud Integration Endpoints

GET /api/v1/integrations/truckercloud/vehicles — Fleet Vehicles Master Registry

Function & Purpose: *Returns all active fleet vehicles mapped with validated 17-character VIN, device IMEI, assigned driver ID, and ECU odometer.*
TruckerCloud uses this endpoint to synchronize vehicle assets, discover newly installed DashCams, and associate telematics.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key (inurum-tc-prod-key-2026)
page	*Query*	Optional. Zero-based page number (default: 0)
size	*Query*	Optional. Number of items per page (default: 50, max: 200)

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "plateNumber": "TRK-9001", "label": "Heavy Hauler 9001", "model": "Kenworth T680", "year": 2024, "assignedDeviceId": "868205040001001", "assignedDriverId": "drv-f4685a30", "assignedDriverName": "Marcus Vance", "odometer": { "value": 145230.8, "unit": "km", "source": "ECU" }, "active": true } ], "page": 0, "size": 50, "totalItems": 1, "totalPages": 1, "last": true }
```

GET /api/v1/integrations/truckercloud/vehicles/{vin} — Vehicle Lookup by 17-Char VIN

Function & Purpose: *Fetches individual vehicle details using its unique 17-character VIN identifier according to ISO 3779 standard.*
Allows TruckerCloud systems to immediately resolve a physical DashCam and current driver assignment from a known chassis VIN.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
vin	*Path*	Required. Validated 17-character vehicle identification number (e.g., 1HD1KAE12FB099999)

Real Response Payload (Sanitized Sample):


```
{ "id": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "plateNumber": "TRK-9001", "label": "Heavy Hauler 9001",
"model": "Kenworth T680", "year": 2024, "assignedDeviceId": "868205040001001", "assignedDriverId":
"drv-f4685a30", "assignedDriverName": "Marcus Vance", "odometer": { "value": 145230.8, "unit": "km", "source":
"ECU" }, "active": true }
```


GET /api/v1/integrations/truckercloud/drivers — Fleet Drivers Directory

Function & Purpose: *Provides a paginated list of all active fleet drivers with CDL license information and current vehicle pairings.*
Used by TruckerCloud to attribute driver safety scores, incident investigations, and ELD compliance records to specific drivers.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
page	*Query*	Optional. Page number (default: 0)
size	*Query*	Optional. Page size (default: 50)

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "drv-f4685a30", "name": "Marcus Vance", "phone": "+1-404-555-0188", "email":
"marcus.vance@inurumfleet.com", "licenseNumber": "GA-CDL-882910", "licenseState": "GA", "assignedVehicleId":
"veh-aebf556b", "assignedVehiclePlate": "TRK-9001", "active": true } ], "page": 0, "size": 50, "totalItems": 1,
"totalPages": 1, "last": true }
```


GET /api/v1/integrations/truckercloud/telemetry/latest — Latest Live GPS & Telematics Fix

Function & Purpose: *Returns the most recent live location, speed, bearing, ignition status, and ECU odometer for all active fleet vehicles.*
Supplies TruckerCloud live dispatch maps with real-time location telemetry received over physical JT/T 808 0x0200 packets.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
terminalId	*Query*	Optional. Filter by specific DashCam IMEI / terminal ID

Real Response Payload (Sanitized Sample):


```
[ { "terminalId": "868205040001001", "vehicleId": "veh-aebf556b", "plateNumber": "TRK-9001", "vin":
"1HD1KAE12FB099999", "lat": 33.7490, "lng": -84.3880, "speed": 62.4, "heading": 185, "ignition": true,
"odometer": { "value": 145230.8, "unit": "km", "source": "ECU" }, "timestamp": "2026-11-03T05:30:12Z" } ]
```

GET /api/v1/integrations/truckercloud/telemetry/history — Historical GPS Breadcrumbs & Trajectory

Function & Purpose: *Retrieves historical GPS position records supporting vehicle/terminal filtering, start/end UTC dates, and pagination.*
Allows TruckerCloud to perform exact historical route playback, geofence ingress/egress audits, and stop verification.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
vehicleId	*Query*	Optional. Filter by vehicle unique ID (e.g., veh-aebf556b)
terminalId	*Query*	Optional. Filter by DashCam IMEI
startTime	*Query*	Optional. UTC ISO 8601 start time (e.g., 2026-11-03T00:00:00Z)
endTime	*Query*	Optional. UTC ISO 8601 end time (e.g., 2026-11-03T12:00:00Z)
page / size	*Query*	Optional. Pagination parameters (size max: 1000)

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "pos-882193", "terminalId": "868205040001001", "vehicleId": "veh-aebf556b", "vin":
"1HD1KAE12FB099999", "latitude": 33.7490, "longitude": -84.3880, "altitude": 310.5, "speed": 62.4, "heading":
185, "ignition": true, "odometerKm": 145230.8, "odometerSource": "ECU", "timestamp": "2026-11-03T05:30:12Z" } ],
"page": 0, "size": 100, "totalItems": 1, "totalPages": 1, "last": true }
```


GET /api/v1/integrations/truckercloud/safety/events — ADAS, DMS & G-Sensor Safety Events

Function & Purpose: *Returns camera-generated safety incidents including harsh braking, rapid acceleration, collision, distraction, and fatigue.*
Includes vehicle and driver association, exact GPS location at time of event, severity score, and video proof availability.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
type	*Query*	Optional. Event type (e.g., HARSH_BRAKE, COLLISION, DISTRACTION)
severity	*Query*	Optional. Severity filter (LOW, MEDIUM, HIGH, CRITICAL)
vehicleId	*Query*	Optional. Filter by vehicle ID
page / size	*Query*	Optional. Pagination controls

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "evt-7729104", "type": "HARSH_BRAKE", "severity": "HIGH", "timestamp":
"2026-11-03T04:15:22Z", "terminalId": "868205040001001", "vehicleId": "veh-aebf556b", "vin":
"1HD1KAE12FB099999", "driverId": "drv-f4685a30", "driverName": "Marcus Vance", "latitude": 33.7550, "longitude":
-84.3900, "speedKmh": 45.2, "gForce": -0.68, "hasVideo": true, "roadCameraUrl":
"http://3.85.81.10:8080/media/incidents/evt-7729104_ch1.m3u8", "cabinCameraUrl":
"http://3.85.81.10:8080/media/incidents/evt-7729104_ch2.m3u8" } ], "page": 0, "size": 50, "totalItems": 1,
"totalPages": 1, "last": true }
```

GET /api/v1/safety/recording-url — Incident Video Proof Playback URLs

Function & Purpose: *Generates secure, documented HLS streaming links for road and cabin camera channels linked to a specific safety event ID.*
Allows TruckerCloud portal users to review high-definition dual-camera video evidence with signed link expiration safeguards.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key (or Bearer JWT)
eventId	*Query*	Required. Safety event identifier (e.g., evt-7729104)

Real Response Payload (Sanitized Sample):


```
{ "eventId": "evt-7729104", "terminalId": "868205040001001", "vehicleId": "veh-aebf556b", "vin":
"1HD1KAE12FB099999", "roadCameraUrl": "http://3.85.81.10:8080/media/incidents/evt-7729104_ch1.m3u8",
"cabinCameraUrl": "http://3.85.81.10:8080/media/incidents/evt-7729104_ch2.m3u8", "mediaStatus": "READY",
"expiresAt": "2026-11-03T16:30:00Z" }
```


GET /api/v1/integrations/truckercloud/trips — Fleet Trips & Mileage Records

Function & Purpose: *Retrieves summarized vehicle trips with departure/arrival timestamps, start/end coordinates, duration, and distance traveled.*
Supplies operational data required by TruckerCloud for fuel consumption estimation, route compliance, and IFTA tax audits.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
vehicleId	*Query*	Optional. Filter trips by vehicle ID
startTime / endTime	*Query*	Optional. Date range filter in UTC ISO 8601
page / size	*Query*	Optional. Pagination parameters

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "trip-5501", "vehicleId": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "driverId":
"drv-f4685a30", "startTime": "2026-11-03T01:00:00Z", "endTime": "2026-11-03T05:00:00Z", "startLocation": {
"lat": 33.7490, "lng": -84.3880, "address": "Atlanta Depot" }, "endLocation": { "lat": 32.0809, "lng": -81.0912,
"address": "Savannah Terminal" }, "distanceKm": 384.6, "durationMinutes": 240, "odometerStartKm": 144846.2,
"odometerEndKm": 145230.8, "odometerSource": "ECU" } ], "page": 0, "size": 50, "totalItems": 1, "totalPages": 1,
"last": true }
```


POST /api/v1/video/sd/query — Remote MicroSD Card Video Query (JT/T 808 0x9205)

Function & Purpose: *Dispatches an over-the-air JT/T 808 0x9205 instruction to the DashCam to inspect recorded video clips stored on its MicroSD card.*
Enables TruckerCloud to discover all available continuous recording blocks without transferring heavy video data to cloud storage.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
deviceId	*Body JSON*	Required. Terminal IMEI / device ID
channel	*Body JSON*	Required. Camera channel: 1 (Road) or 2 (Cabin)
startTime	*Body JSON*	Required. Query start timestamp in UTC ISO 8601
endTime	*Body JSON*	Required. Query end timestamp in UTC ISO 8601

Real Response Payload (Sanitized Sample):


```
{ "success": true, "deviceId": "868205040001001", "channel": 1, "recordings": [ { "channel": 1, "startTime":
"2026-11-03T03:00:00Z", "endTime": "2026-11-03T03:30:00Z", "alarmType": 0, "mediaType": 0, "streamType": 0,
"storageType": 0, "fileSizeBytes": 524288000 } ], "total": 1 }
```

POST /api/v1/video/sd/retrieve — On-Demand Video Clip Retrieval Job (JT/T 808 0x9201)

Function & Purpose: *Creates an asynchronous video retrieval job commanding the DashCam via 0x9201 to upload a specific time clip to cloud storage.*
Guarantees that cloud bandwidth and AWS costs are consumed only for explicitly requested incidents and investigation clips.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
deviceId	*Body JSON*	Required. Terminal IMEI
channel	*Body JSON*	Required. 1 for Road camera, 2 for Cabin camera
startTime	*Body JSON*	Required. UTC ISO 8601 clip start
endTime	*Body JSON*	Required. UTC ISO 8601 clip end

Real Response Payload (Sanitized Sample):


```
{ "jobId": "job-ret-991204", "status": "QUEUED", "deviceId": "868205040001001", "channel": 1, "startTime":
"2026-11-03T03:10:00Z", "endTime": "2026-11-03T03:15:00Z", "message": "Video retrieval job successfully queued
with physical DashCam." }
```


GET /api/v1/video/jobs/{jobId} — Video Retrieval Job Progress & Download URLs

Function & Purpose: *Polls the real-time upload progress and completion status of an on-demand DashCam video extraction job.*
Returns ready HLS playback URLs and MP4 download links once the physical camera completes JT/T 1078 media upload.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
jobId	*Path*	Required. Retrieval job identifier returned by /api/v1/video/sd/retrieve

Real Response Payload (Sanitized Sample):


```
{ "jobId": "job-ret-991204", "status": "COMPLETED", "progressPercent": 100, "deviceId": "868205040001001",
"channel": 1, "playbackUrl": "http://3.85.81.10:8080/media/retrievals/job-ret-991204.m3u8", "downloadUrl":
"http://3.85.81.10:8080/media/retrievals/job-ret-991204.mp4", "completedAt": "2026-11-03T05:35:10Z" }
```


3. Integration Acceptance & Support Verification

Live Production Validation: All endpoints in this specification have been validated against our live AWS EC2 production deployment (3.85.81.10:8080) backed by MySQL 8.4 persistence and active WanWay H20P JT/T 808/1078 gateway services. All 13 core TruckerCloud acceptance tests completed with a 100% pass rate.

Interactive Testing:
• **Live Swagger UI:** http://3.85.81.10:8080/swagger-ui/index.html
• **Raw OpenAPI 3.0 Spec:** http://3.85.81.10:8080/v3/api-docs
• **Postman Collection:** A pre-configured collection file Inurum_TruckerCloud_Postman_Collection.json is provided in the repository with real, sanitized requests and pre-set X-API-KEY variables.

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